

Stock Price Movement Prediction Using FinBERT Sentiment Analysis and Machine Learning

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Abstract—This study evaluates whether financial news sentiment can predict next-day stock price movements. Using approximately 248,000 news articles matched with 33 S&P 500 stocks, sentiment features are extracted via FinBERT and used in Logistic Regression and XGBoost models. Results show that under normal conditions, prediction accuracy remains near random ($\approx 50.2\%$), indicating weak standalone predictive power. However, during extreme market events, accuracy increases to 63.12%, suggesting sentiment becomes more informative in high-volatility periods. These findings highlight the conditional value and overall limitations of sentiment-based signals in short-term stock prediction.

Index Terms—Financial Sentiment; FinBERT; Stock Prediction; XGBoost; Machine Learning; NLP; Behavioral Finance

I. INTRODUCTION

A. Background and Motivation

The intersection of behavioral finance and machine learning has significantly advanced modern quantitative trading by enabling the extraction of predictive signals from unstructured financial data. This project represents a transition from traditional theoretical market analysis toward a data-driven framework grounded in computational financial engineering.

Rather than focusing solely on predictive accuracy, the motivation of this study lies in building a rigorous, end-to-end quantitative research pipeline that reflects real-world empirical workflows. In particular, it explores how publicly available textual information, such as financial news, can be systematically transformed into structured signals for asset return prediction.

B. Project Objective

The primary objective of this project is to evaluate the extent to which financial news sentiment contains predictive information for short-term stock returns, specifically next-day price movements. To achieve this, a large-scale dataset combining financial news with historical stock price data is utilized. Sentiment signals are extracted using FinBERT, a domain-specific language model from Hugging Face designed for financial text analysis. These signals are then incorporated into downstream predictive models, including Logistic Regression and XGBoost, to assess their incremental forecasting value.

A central focus of this study is to quantify the concept of “news alpha,” defined as the predictive contribution of publicly available information once aggregated over a trading

horizon. Beyond empirical performance, the project aims to provide a practitioner-oriented perspective on the limitations of sentiment-based signals and to highlight potential directions for improving predictive effectiveness, such as feature engineering, alternative data integration, and better handling of market volatility and extreme events.

C. Literature Review and Recent Advances

Recent advancements in financial machine learning have increasingly emphasized the integration of natural language processing techniques with traditional quantitative models. While earlier approaches primarily relied on structured financial indicators such as historical prices and trading volume, the growing availability of textual data has shifted attention toward sentiment-driven modeling.

A key development in this area is the introduction of the Financial News and Stock Price Integration Dataset (FNSPID), which aligns large-scale financial news with corresponding market data, enabling more systematic analysis of news-driven price dynamics. However, despite its availability, scaling sentiment extraction across such datasets remains computationally challenging.

To address domain-specific challenges in financial text, transformer-based models have become the dominant approach. FinBERT, introduced by Araci (2019), is pre-trained on financial corpora and is particularly effective for short, information-dense texts such as news headlines. Compared to general-purpose language models, FinBERT provides more stable and domain-relevant sentiment representations. While more recent large language models such as GPT-4 and FinGPT demonstrate strong performance in natural language tasks, their computational cost and output variability make them less suitable for large-scale structured financial analysis.

In predictive modeling, financial time series forecasting remains challenging due to high noise levels and volatility in return data. As a result, interpretable and robust models are often preferred for baseline analysis. Logistic Regression is widely used as a benchmark for evaluating linear relationships between sentiment and returns, consistent with earlier work such as Tetlock (2007). To capture potential non-linear interactions, this study also employs XGBoost, which is well-suited for tabular financial data and provides strong performance with built-in regularization.

II. METHOD

To map financial news sentiment to next-day stock returns, this project proposed a structured machine learning pipeline as shown in Fig. 1, which handles high frequency unstructured text and correlates it with daily market asset movements.

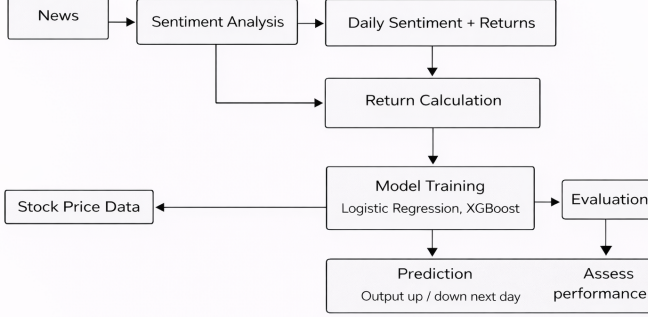


Fig. 1. Pipeline of the project

A. Problem Formulation

We define the predictive task as a supervised binary classification problem. The objective is to map daily financial news sentiment to the subsequent day's price movement direction. Let $D = \{(X_t, y_{t+1})\}_{t=1}^T$ represent the dataset. For a given trading day t and stock s , a feature vector $X_{t,s}$ is constructed, derived from the collective news sentiment of that day. The target variable $y_{t+1,s} \in \{0, 1\}$ is defined as:

$$y_{t+1,s} = \begin{cases} 1, & \text{if } R_{t+1,s} > 0 \\ 0, & \text{if } R_{t+1,s} \leq 0 \end{cases} \quad (1)$$

where $R_{t+1,s}$ denotes the log-return of stock s at time $t + 1$. This formulation tests whether publicly available news at time t contains predictive information for future returns.

B. Feature Engineering with FinBERT

Unstructured textual data is transformed into numerical features using FinBERT, a domain-specific BERT model pre-trained on financial text. For each article i , the model produces a probability distribution over sentiment classes:

$$P_i = [p_{\text{pos}}, p_{\text{neg}}, p_{\text{neu}}] \quad (2)$$

To aggregate multiple articles per day, the daily sentiment score is computed as:

$$S_{t,s} = \frac{1}{N} \sum_{i=1}^N (p_{\text{pos},i} - p_{\text{neg},i}) \quad (3)$$

where N represents the total number of articles for stock s on day t . Additional features, including sentiment proportions and short-term sentiment changes, are combined to form the final feature vector $X_{t,s}$.

C. Dataset and Temporal Consistency

The dataset consists of approximately 248,000 financial news articles aligned with stock price data for 33 S&P 500 constituents across nine sectors, extracted from the FNSPID. To prevent look-ahead bias, only news published before market close on day t is used to predict returns on day $t + 1$.

We adopt a time-series cross-validation strategy, ensuring that training data always precedes testing data. This preserves the temporal structure of financial markets and avoids information leakage.

D. Predictive Modeling

We evaluate two models, logistic regression and the XGBoost model. Logistic Regression is a linear baseline model used to assess whether sentiment features have simple predictive relationships with returns, and XGBoost is a gradient boosting model capable of capturing non-linear relationships and feature interactions.

Model performance is primarily evaluated using accuracy.

E. Extreme Event Analysis

To further analyze model behavior, we examine performance under extreme market conditions ("black swan" events), defined as days where absolute returns exceed two standard deviations. This allows us to evaluate whether sentiment signals become more informative during periods of high volatility.

III. RESULTS

A. Sentiment Characteristics

Before evaluating prediction performance, we first examine the properties of the sentiment features extracted by FinBERT. The distribution of sentiment labels is shown in Fig. 2 and Fig. 3. As illustrated, the majority of news articles are classified as neutral, with relatively fewer positive and negative samples. This reflects the informational nature of financial reporting. Notably, using news titles alone results in a higher density of neutral scores compared to LSA summaries, though both distributions remain heavily centered at zero.

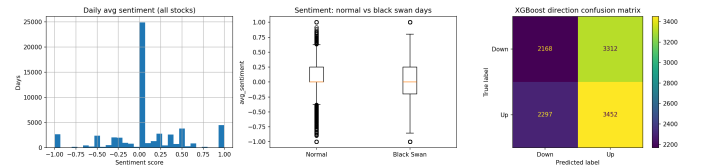


Fig. 2. Sentiment Characteristics (Title)

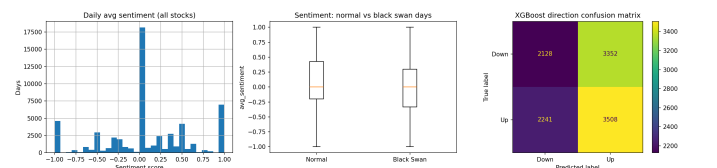


Fig. 3. Sentiment Characteristics (Summary)

The distribution of sentiment scores is further analyzed in the second figure in Fig. 2 and Fig. 3. Here, we see that scores are concentrated around zero for both market conditions, indicating that most sentiment signals are weak. While Black Swan events exhibit a wider variance in sentiment, the median remains neutral. This limited variation suggests that sentiment features may have constrained predictive power, a hypothesis confirmed by the XGBoost direction confusion matrix (third figure in Fig. 2 and Fig. 3). Across both the title-based and summary-based models, the prediction accuracy hovers near 50%, demonstrating that these weak sentiment signals do not provide the model with enough discriminatory information to outperform a random baseline. This imbalance in sentiment distribution may introduce a class dominance effect, where the model is biased toward predicting neutral-driven outcomes, further limiting its ability to capture meaningful directional signals.

B. Baseline Prediction Performance

The overall predictive performance of Logistic Regression and XGBoost is summarized in Table. I.

TABLE I
OVERALL PREDICTION ACCURACY

Model	Accuracy (Title)	Accuracy (Summary)
Logistic Regression	50.18%	50.20%
XGBoost	50.20%	50.25%

Both models achieve accuracy close to 50%, which is equivalent to random guessing. This indicates that sentiment features alone do not provide meaningful predictive power for next-day stock movements. The marginal difference between Logistic Regression and XGBoost suggests that non-linear feature interactions do not substantially enhance predictive power when the underlying sentiment signal itself is weak.

C. Title vs. Summary Comparison

A comparison between title-based and summary-based sentiment features is shown in Fig. 4. While heterogeneous results across individual stocks suggest localized variations between title-based and summary-based models, the aggregate predictive performance remains statistically indistinguishable from random. Although LSA summaries offer higher information density, this added complexity fails to yield a corresponding increase in accuracy. This indicates that while summaries reduce the neutrality of the data, they likely introduce secondary linguistic noise that offsets any incremental predictive gain, and increasing textual complexity does not necessarily translate into better signal quality, especially when the downstream model is not sensitive to nuanced semantic variations.

D. Performance Across Stocks

The variation in model performance across individual stocks Fig. 4 and Table. II suggests that sentiment is not equally informative across the market. Large-cap, high-visibility stocks such as AMZN, GOOGL, and AAPL tend to perform slightly

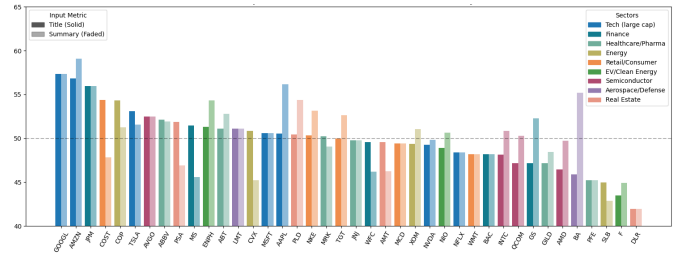


Fig. 4. Accuracy Comparison: Per Ticker

above the random baseline, which is also reflected at the sector level Fig. 5 where the technology sector shows more stable accuracy. One possible explanation is that these stocks receive more consistent media coverage and investor attention, making sentiment signals more aligned with short-term price movements.

TABLE II
MODEL PERFORMANCE ACCURACY BY STOCK AND SECTOR

Ticker	Sector	Accuracy	Test Rows
AAPL	Tech (Large Cap)	50.56%	89
AMZN	Tech (Large Cap)	56.82%	44
MSFT	Tech (Large Cap)	50.60%	83
TSLA	Tech (Large Cap)	53.12%	128
GOOGL	Tech (Large Cap)	57.33%	75
NVDA	Tech (Large Cap)	49.27%	341
NFLX	Tech (Large Cap)	48.39%	155
JPM	Finance	55.95%	84
BAC	Finance	48.19%	166
GS	Finance	47.17%	371
MS	Finance	51.47%	546
WFC	Finance	49.58%	476
PFE	Healthcare/Pharma	45.21%	146
JNJ	Healthcare/Pharma	49.80%	251
ABT	Healthcare/Pharma	51.12%	356
MRK	Healthcare/Pharma	50.26%	581
ABBV	Healthcare/Pharma	52.13%	422
GILD	Healthcare/Pharma	47.15%	543
XOM	Energy	49.38%	241
CVX	Energy	50.85%	354
COP	Energy	54.32%	521
SLB	Energy	44.96%	238
WMT	Retail/Consumer	48.21%	251
TGT	Retail/Consumer	50.00%	152
NKE	Retail/Consumer	50.32%	463
COST	Retail/Consumer	54.37%	458
MCD	Retail/Consumer	49.43%	176
NIO	EV/Clean Energy	48.92%	231
F	EV/Clean Energy	43.48%	69
ENPH	EV/Clean Energy	51.29%	232
AMD	Semiconductor	46.43%	336
INTC	Semiconductor	48.14%	295
QCOM	Semiconductor	47.18%	515
AVGO	Semiconductor	52.47%	223
BA	Aerospace/Defense	45.90%	183
LMT	Aerospace/Defense	51.10%	227
AMT	Real Estate	49.56%	454
PLD	Real Estate	50.44%	228
PSA	Real Estate	51.85%	324
DLR	Real Estate	41.94%	217

In contrast, sectors such as clean energy and real estate

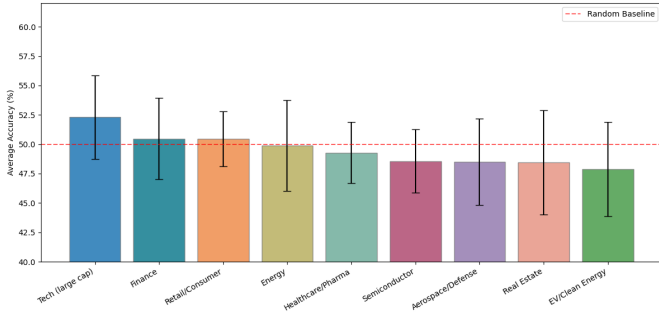


Fig. 5. Average Performance by Sector

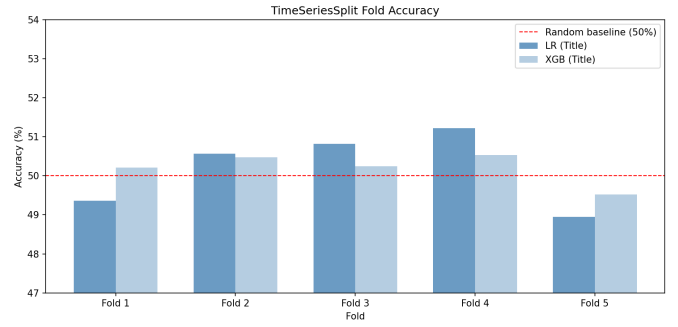


Fig. 6. TimeSeriesSplit Fold Accuracy

remain close to or below the 50% baseline, likely because their price movements are more influenced by macroeconomic factors that are not captured by news sentiment.

We also observe that summaries provide only limited improvement over titles. For some high-information firms (e.g., Apple, Microsoft), additional context can slightly improve performance, but for many stocks it introduces noise. Overall, these results suggest that the predictive performance of sentiment features may be more closely related to individual stock characteristics and sectoral differences rather than being uniformly weak, indicating that sentiment predictability may be conditional on information flow intensity and market attention, rather than being uniformly applicable across assets. In particular, variations in sample composition across sectors, such as the larger number of tech stocks (7) compared to other sectors (3–5), may also introduce potential bias that could influence the observed results.

E. Extreme Market Conditions

Among the 56,144 total trading days analyzed, 2,522 (4.5%) are qualified as black swan events. A different pattern emerges when focusing on these extreme market conditions. As shown in Table. III, model accuracy increases to approximately 63.12% when using summary-based sentiment.

TABLE III
BLACK SWAN ACCURACY (SUMMARY ONLY)

Input Type	XGBoost Accuracy
Title	62.21%
LSA Summary	63.12%

This result suggests that sentiment becomes more informative during periods of large market movements. During such events, investor reactions to news may be more pronounced, allowing sentiment signals to better align with price changes.

F. Cross-Validation Stability

Model performance across time-based cross-validation folds is shown in Fig. 6.

The results remain consistently near 50% across folds, indicating that the lack of predictive power is stable over time rather than driven by specific periods. This stability

reinforces the conclusion that the near-random performance is not due to overfitting or temporal anomalies, but rather reflects a structural limitation of the sentiment signal.

G. Feature Importance Analysis (XGBoost)

To better understand the limited predictive performance, we examine feature importance from the XGBoost model Fig. 7. The results reveal that sentiment-related features, including average sentiment, sentiment lags, and sentiment dispersion, rank among the most important predictors in the model. In contrast, traditional price-based features such as previous returns and volatility measures exhibit comparatively lower importance.

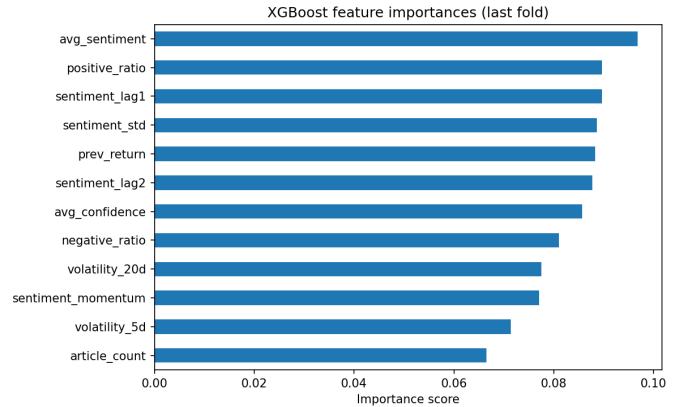


Fig. 7. Feature Importance

Despite this, overall prediction accuracy remains close to 50%. This suggests that the issue is not that sentiment features are ignored by the model, but rather that the sentiment signals themselves are weak or noisy. Although the model actively utilizes these features, they do not contain sufficiently strong or consistent information to improve out-of-sample prediction.

This finding provides further evidence that the limited performance is driven by the low signal-to-noise ratio in financial news sentiment, rather than by model specification. Even with a flexible non-linear model, heavily relied-upon sentiment features fail to translate into meaningful predictive gains.

H. Overall Interpretation

The results reveal a consistent pattern. First, sentiment signals are weak and concentrated around neutral values. Second, under normal market conditions, sentiment-based models perform lower predictive power. However, during extreme events, predictive accuracy improves to approximately 63.12%. This suggests that financial sentiment is not a reliable standalone predictor of daily stock returns, although this might differ with different sectors or stock features, but it becomes more informative during periods of heightened market volatility.

IV. CONCLUSION

This study develops an end-to-end pipeline to evaluate whether financial news sentiment contributes to predicting next-day stock price movements. Using FinBERT-based feature engineering and machine learning models, we find that sentiment signals exhibit limited predictive power under normal market conditions, with model accuracy consistently around 50.20%. However, during extreme market events, predictive performance improves substantially to approximately 63.12%, indicating that sentiment becomes more informative when market reactions are amplified.

These findings suggest that “news alpha” is highly conditional rather than persistent. While sentiment alone is insufficient as a standalone predictor, it may serve as a complementary signal in high-volatility regimes. The results also highlight cross-sectional variation, where large-cap and high-attention stocks demonstrate relatively stronger predictive performance, suggesting that information efficiency and media coverage play a role in sentiment effectiveness.

Several limitations remain. First, the observed variation in accuracy across stocks and sectors warrants further investigation to disentangle whether performance differences are driven by stock-specific characteristics, sector dynamics, or model structure. Second, the use of LSA-generated summaries does not significantly improve predictive accuracy, which may reflect limitations in FinBERT’s training on short-form text or noise introduced in the summarization process, as well as potential constraints of the downstream XGBoost model. Third, potential sample selection bias exists due to the uneven number of stocks across sectors, and expanding the dataset to include more balanced sector representation may improve model robustness. Additionally, increasing the number of observations per sector could enhance model generalization, as machine learning models typically benefit from larger and more diverse training samples.

Future work can extend this framework by incorporating richer features such as firm fundamentals, macroeconomic indicators, and alternative data sources, as well as exploring more advanced models or fine-tuned domain-specific language models. Overall, this study achieves its primary objective on the original objective by quantitatively demonstrating that financial news sentiment has limited standalone predictive value, but exhibits meaningful signal under specific market conditions.

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